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Lab 1: Implementation of Source code management using git commands through GitHub

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OBJECTIVE

This lab explores how Git and GitHub can be used together to manage source code effectively, from tracking changes to collaborating with a team.

AIM

To gain a working understanding of Source Code Management (SCM) by using Git and GitHub to track changes, manage different versions of a project, and collaborate on software development.

THEORY

Source Code Management (SCM) refers to the practice of monitoring and controlling changes made to a codebase over time. It allows developers to collaborate more smoothly, keep a clear history of every version, and roll back to earlier states of the project whenever needed.

Git is a distributed version control system that lets developers log changes to their files and stay in sync with the rest of their team, even when working independently.

GitHub, meanwhile, is a cloud-based platform built around Git. It goes beyond simple repository hosting by offering tools for collaboration, including pull requests, issue tracking, and project boards.

Together, Git and GitHub form the backbone of Agile Software Development, as they make it easier for teams to work in parallel, integrate their work continuously, keep track of every version, and review each other's code.

OBJECTIVES

1. Understand the fundamentals of Git.
2. Create and manage Git repositories.
3. Track changes to a project through commits.
4. Link a local repository to GitHub.
5. Push and pull updates between local and remote repositories.
6. Build a practical understanding of collaborative software development.

ALGORITHM

- Install Git on the local system.
- Create a GitHub account.
- Set up a new repository on GitHub.
- Initialize a Git repository locally.

- Add the project files to the repository.
- Stage the files using Git.
- Commit the staged changes.
- Link the local repository to the GitHub repository.
- Push the committed files to GitHub.
- Confirm that everything has synced correctly.

CODE

Initialize Repository

```
git init
```

Configure User Information

```
git config --global user.name "your name"
git config --global user.email "your@email.com"
```

Check Repository Status

```
git status
```

Add Files

```
git add .
```

Commit Changes

```
git commit -m "Initial Commit"
```

Connect Repository to GitHub

```
git remote add origin https://github.com/username/repository.git
```

Push to GitHub

```
git branch -M main
git push -u origin main
```

Pull Latest Changes

```
git pull origin main
```

OUTPUT

The Git repository was set up and connected to GitHub without any issues:

7. The repository was created successfully.
8. Project files were being tracked by Git.
9. Changes were committed as expected.
10. The remote repository was linked correctly.
11. The project was pushed to GitHub successfully.

RESULT

Source Code Management using Git and GitHub was implemented successfully. Core version control operations, including repository creation, staging, committing, and syncing with GitHub, were all carried out without errors.

CONCLUSION

This lab offered hands-on experience with Source Code Management through Git and GitHub. It showed how version control systems make software projects easier to manage by preserving code history, enabling smoother collaboration, and supporting the practices central to Agile Software Development.